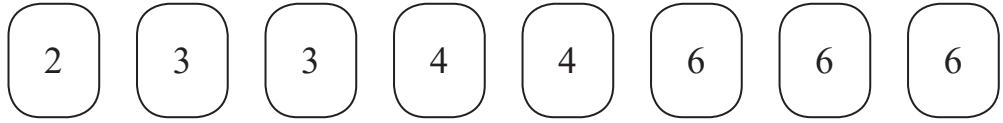
16 Here are 8 cards.

There is a number on each card.



Erin puts the 8 cards in a bag.
She takes at random a card from the bag and does not replace it.
Erin then takes at random a second card from the bag.

Calculate the probability that the number on the second card is double the number on the first card.

Only successful pairs are (2,4) and (3,6).

Without replacement:

P(2 then4) = (1/8) × (2/7) = 2/56.

1 mark
One valid ordered pair

P(3 then6) = (2/8) × (3/7) = 6/56.

P(total) = 2/56 + 6/56 = 8/56 = 1/7.

1 mark
Add both probabilities

1 mark
1/7

(Total for Question 16 is 3 marks)